

Chiamanda Ononiwu

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EDUCATION

McNeese State University

Bachelor of Science in Electrical Engineering, minor in Computer Science; **GPA: 4.0**

Lake Charles, LA

Graduation date: May 2028

Awards: Dean's List 2024 – Present, McNeese Honors Scholar

Cornell University, Machine Learning Foundations e-Certificate, Break Through Tech

May 2025 – August 2025

TECHNICAL SKILLS

Languages: Python, C++

AI/ML: PyTorch, TensorFlow, Scikit-learn, LightGBM, CNNs, Vision Transformers, NLP, RAG pipelines, LLMs, Prompt Engineering

Tools & Frameworks: Pandas, NumPy, Matplotlib, SQLite, Git, Gemini API, pdfplumber, pytest, ESP32, Fusion 360, LaTeX

PROFESSIONAL EXPERIENCE

Ursa Space Systems, *Break Through Tech AI fellow*

August 2025 – December 2025

- Developed a **neural network classification system** and extracted **embeddings** from preprocessed data with **vision transformer models** using **PyTorch** to distinguish vessels from icebergs in synthetic aperture radar (SAR) imagery, achieving **80% recall** on a dataset of 1400+ samples.
- Implemented exploratory data analysis (**EDA**) involving **data preprocessing pipelines and feature extraction algorithms** using Python, **SARPy**, and **scikit-image**, improving classification recall by **52.83%**.
- Performed comparative analysis across different model architectures to identify the most reliable model for real-world deployment and communicated findings to support collaborative engineering and research workflows.

CEAMLS 2025 Summer AI Research Institute, *Undergraduate Research Project Manager*

May 2025 – August 2025

- Architected the **full IoT data pipeline** for an AI-powered smart waste monitoring system, integrating **ESP32 sensor hardware** with a Python ML backend; conducted **literature reviews** on IoT-ML systems to inform system design.
- Designed and implemented **data preprocessing algorithms** in Python using **Pandas, NumPy, and Scikit-learn**, improving feature extraction and enabling accurate prediction of waste bin fill time.
- Trained a **Random Forest model** to predict waste collection schedules, improving route efficiency by **40%**; authored the **methodology section** of the IEEE manuscript (In review).

Publications: Ononiwu C., Adike N., Nwankwo K., Ifesanmi O., & Efe S. "AI-Powered Smart Waste Management System for Efficient Urban Cleanliness", IEEE Conference on Visualization & Visual Analytics, Nov 2026 (In review)

PROJECTS

Emotion Responsive AI Robot (ESP32 + Gemini API) [GitHub](#)

Feb 2026 – Present

- Built an **end-to-end voice-to-emotion AI pipeline** on an **ESP32-S3 microcontroller** that captures voice input (INMP441 mic), transmits audio to the **Gemini API** over Wi-Fi, and renders real-time facial expressions on a display, enabling human-like robot interaction.
- Trained a **custom NLP sentiment analysis model** to classify **Gemini's text responses** into 5 emotional states, **bridging NLP and embedded hardware** to advance emotionally intelligent HRI.
- Architected a **full-stack embedded AI system** integrating audio capture, **LLM inference**, **sentiment classification**, graphical display rendering, and TTS audio output.

StudyBot – AI RAG Study Assistant (CodePath) [GitHub](#)

April 2026

- Designed and built an **AI-powered RAG study assistant** that ingests PDF, indexes them via an inverted keyword index, retrieves the top 3 most relevant chunks per query, and uses **Gemini** to answer questions grounded strictly in the document, democratizing personalized study support.
- Engineered **LLM guardrails via prompt engineering** to prevent hallucination, validated with **pytest unit tests**; continuing to improve chunking and retrieval strategies.

LEADERSHIP EXPERIENCE

McNeese NSBE, *TORCH Chair*

April 2025 – Present

- Organized a community **STEM academic tech bowl serving 50+ attendees**, collaborating with Louisiana Region V STEM center and managing on-site logistics.
- Built a centralized database of **15+ STEM outreach partners** across Calcasieu Parish, creating new partnerships that enabled members to deliver STEM lessons to K–12 students.
- Organized trips to different high schools to talk about engineering and college life, leading to some kids deciding to pursue engineering.